

## PROJECT REPORT ON:

# “Bioconductor-Based Sequence and Genomic Annotation Analysis of Cancer-Associated Genes TP53 and EGFR Using R”

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## NUMBER (B) : TP53 — The "Guardian of the Genome"

| Field                | Value             |
|----------------------|-------------------|
| Gene symbol          | TP53              |
| NCBI accession       | NM_000546         |
| Chromosome           | chr17             |
| Start – End (GRCh38) | 7668402 – 7687550 |
| Strand               | -                 |

## Questions

Q1) Download the TP53 sequence using `entrez_fetch()` and save it as "tp53.fasta". How many bases long is the sequence?

Result:

Code:

```
library(Biostrings)
library(GenomicRanges)
library(AnnotationDbi)
library(org.Hs.eg.db)
library(rentrez)

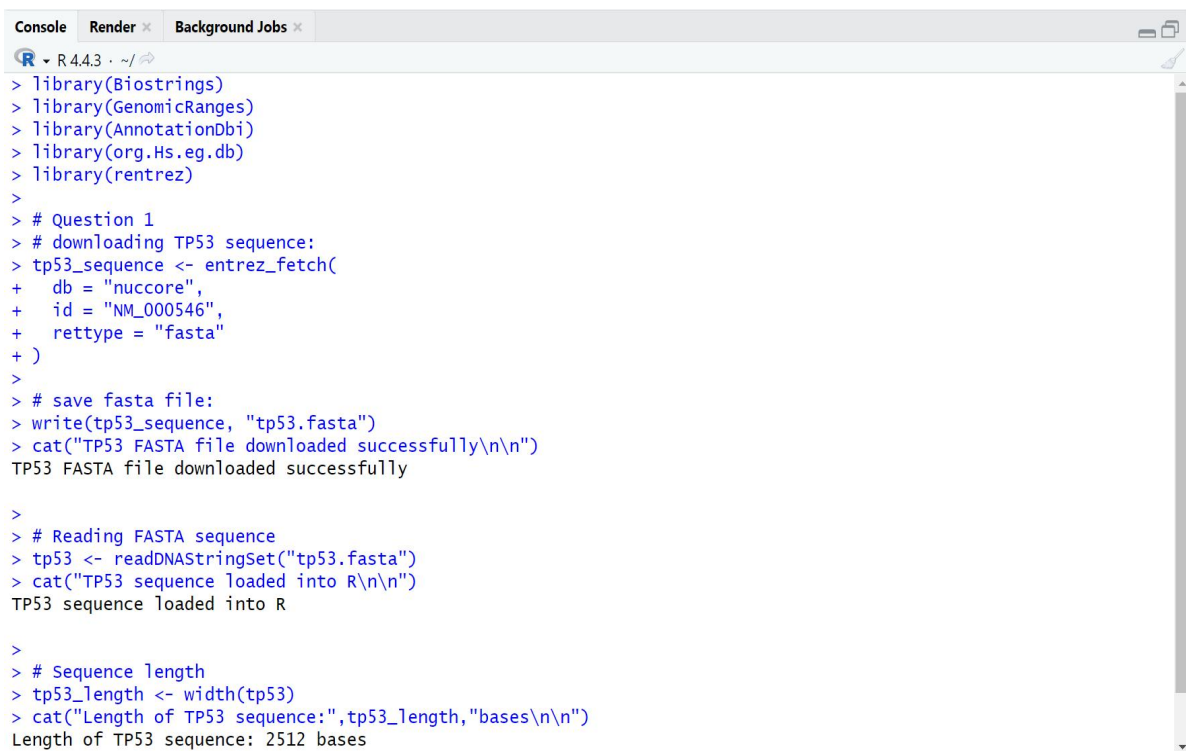
# Question 1
# downloading TP53 sequence:
tp53_sequence <- entrez_fetch(
  db = "nuccore",
  id = "NM_000546",
  rettype = "fasta"
)

# save fasta file:
write(tp53_sequence, "tp53.fasta")
cat("TP53 FASTA file downloaded successfully\n\n")

# Reading FASTA sequence
tp53 <- readDNAStringSet("tp53.fasta")
cat("TP53 sequence loaded into R\n\n")

# Sequence length
tp53_length <- width(tp53)
cat("Length of TP53 sequence:", tp53_length, "bases\n\n")
```

Output:



```
R 4.4.3 · ~/
> library(Biostrings)
> library(GenomicRanges)
> library(AnnotationDbi)
> library(org.Hs.eg.db)
> library(rentrez)
>
> # Question 1
> # downloading TP53 sequence:
> tp53_sequence <- entrez_fetch(
+   db = "nuccore",
+   id = "NM_000546",
+   rettype = "fasta"
+ )
>
> # save fasta file:
> write(tp53_sequence, "tp53.fasta")
> cat("TP53 FASTA file downloaded successfully\n\n")
TP53 FASTA file downloaded successfully
>
> # Reading FASTA sequence
> tp53 <- readDNAStringSet("tp53.fasta")
> cat("TP53 sequence loaded into R\n\n")
TP53 sequence loaded into R
>
> # Sequence length
> tp53_length <- width(tp53)
> cat("Length of TP53 sequence:", tp53_length, "bases\n\n")
Length of TP53 sequence: 2512 bases
```

**Q2) What are the first 30 bases of the sequence?**

**Result:**

**Code:**

```
# Question 2:
# First 30 nucleotide bases
first30 <- subseq(
  tp53,
  start = 1,
  end = 30
)

cat("First 30 bases of TP53:", as.character(first30), "\n\n")
```

**Output:**

A screenshot of an R console window showing the execution of R code. The console has tabs for 'Console', 'Render', and 'Background Jobs'. The output shows the first 30 bases of the TP53 sequence: 'CTCAAAAGTCTAGAGCCACCGTCCAGGGAG'.

```
R 4.4.3 · ~/
> # Question 2:
> # First 30 nucleotide bases
> first30 <- subseq(
+   tp53,
+   start = 1,
+   end = 30
+ )
>
> cat("First 30 bases of TP53:", as.character(first30), "\n\n")
First 30 bases of TP53: CTCAAAAGTCTAGAGCCACCGTCCAGGGAG
> |
```

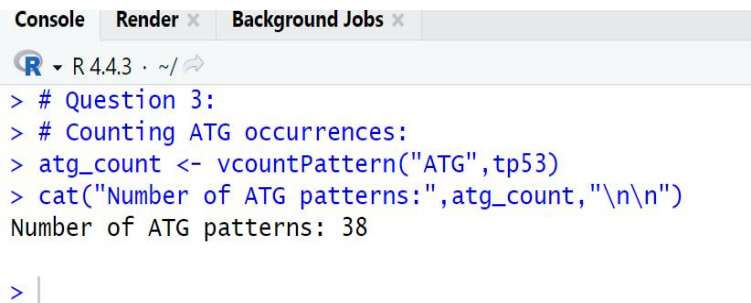
**Q3) How many times does the pattern "ATG" appear in the sequence?**

**Result:**

**Code:**

```
# Question 3:
# Counting ATG occurrences:
atg_count <- vcountPattern("ATG",tp53)
cat("Number of ATG patterns:",atg_count,"\n\n")
```

**Output:**

A screenshot of an R console window showing the execution of R code. The console has tabs for 'Console', 'Render', and 'Background Jobs'. The output shows the number of ATG patterns in the sequence: 38.

```
R 4.4.3 · ~/
> # Question 3:
> # Counting ATG occurrences:
> atg_count <- vcountPattern("ATG",tp53)
> cat("Number of ATG patterns:",atg_count,"\n\n")
Number of ATG patterns: 38
> |
```

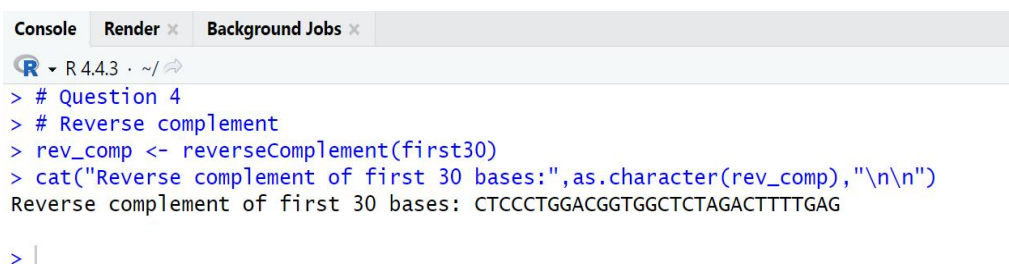
**Q4) What is the reverse complement of the first 30 bases?**

**Result:**

**Code:**

```
# Question 4
# Reverse complement
rev_comp <- reverseComplement(first30)
cat("Reverse complement of first 30 bases:",as.character(rev_comp),"\n\n")
```

**Output:**

A screenshot of an R console window showing the execution of R code. The console has tabs for 'Console', 'Render', and 'Background Jobs'. The output shows the reverse complement of the first 30 bases: 'CTCCCTGGACGGTGGCTCTAGACTTTTGAG'.

```
R 4.4.3 · ~/
> # Question 4
> # Reverse complement
> rev_comp <- reverseComplement(first30)
> cat("Reverse complement of first 30 bases:",as.character(rev_comp),"\n\n")
Reverse complement of first 30 bases: CTCCCTGGACGGTGGCTCTAGACTTTTGAG
> |
```

**Q5) Create a GRanges object for TP53 using the table above. What is its width (end – start)?**

**Result:**

**Code:**

```
# Question 5
# Creating GRanges object for TP53
tp53_range <- GRanges(
  seqnames = "chr17",

  ranges = IRanges(
    start = 7668402,
    end = 7687550
  ),

  strand = "-"
)

print(tp53_range)
```

**Output:**

```
Console Render × Background Jobs ×
R 4.4.3 · ~/
> # Question 5
> # Creating GRanges object for TP53
> tp53_range <- GRanges(
+   seqnames = "chr17",
+   ranges = IRanges(
+     start = 7668402,
+     end = 7687550
+   ),
+   strand = "-"
+ )
> print(tp53_range)
GRanges object with 1 range and 0 metadata columns:
      seqnames      ranges strand
      <Rle>        <IRanges> <Rle>
[1] chr17 7668402-7687550 -
-----
seqinfo: 1 sequence from an unspecified genome; no seqlengths
>
> cat("\nGenomic width:", width(tp53_range), "bases\n\n")
Genomic width: 19149 bases
>
```

**Q6) Using AnnotationDbi and org.Hs.eg.db, what is the ENTREZ ID and full gene name for "TP53"?**

**Result:**

**Code:**

```
# TP53 annotation
tp53_annotation <- select(
  org.Hs.eg.db,

  keys = "TP53",

  columns = c(
    "ENTREZID",
    "GENENAME"
  ),

  keytype = "SYMBOL"
)

print(tp53_annotation)
```

**Output:**

```
Console Render × Background Jobs ×
R 4.4.3 · ~/
> # Question 6:
> # TP53 annotation
> tp53_annotation <- select(
+   org.Hs.eg.db,
+   keys = "TP53",
+   columns = c(
+     "ENTREZID",
+     "GENENAME"
+   ),
+   keytype = "SYMBOL"
+ )
'select()' returned 1:1 mapping between keys and columns
> print(tp53_annotation)
SYMBOL ENTREZID GENENAME
1 TP53 7157 tumor protein p53
>
```

**Q7) In 2–3 sentences, summarize what you learned about TP53's sequence and location.**

**Result:** TP53 is located on the minus strand of chromosome 17, spanning 19,149 bases (7,668,402–7,687,550, GRCh38). The analyzed TP53 mRNA transcript (NM\_000546) was found to be 2,512 bases long and contained 38 ATG occurrences. Known as the "guardian of the genome," TP53 encodes a tumor suppressor protein involved in maintaining genomic stability by regulating cell cycle control and apoptosis, making it one of the most important genes studied in cancer biology.

## Number (C) : EGFR — Lung Cancer / Targeted Therapy Gene

| Field                | Value               |
|----------------------|---------------------|
| Gene symbol          | EGFR                |
| NCBI accession       | NM_005228           |
| Chromosome           | chr7                |
| Start – End (GRCh38) | 55019017 – 55211628 |
| Strand               | +                   |

### Questions

**Q1) Download the EGFR sequence using `entrez_fetch()` and save it as "egfr.fasta". How many bases long is the sequence?**

**Result:**

```
Code: library(Biostrings)
library(GenomicRanges)
library(AnnotationDbi)
library(org.Hs.eg.db)
library(rentrez)

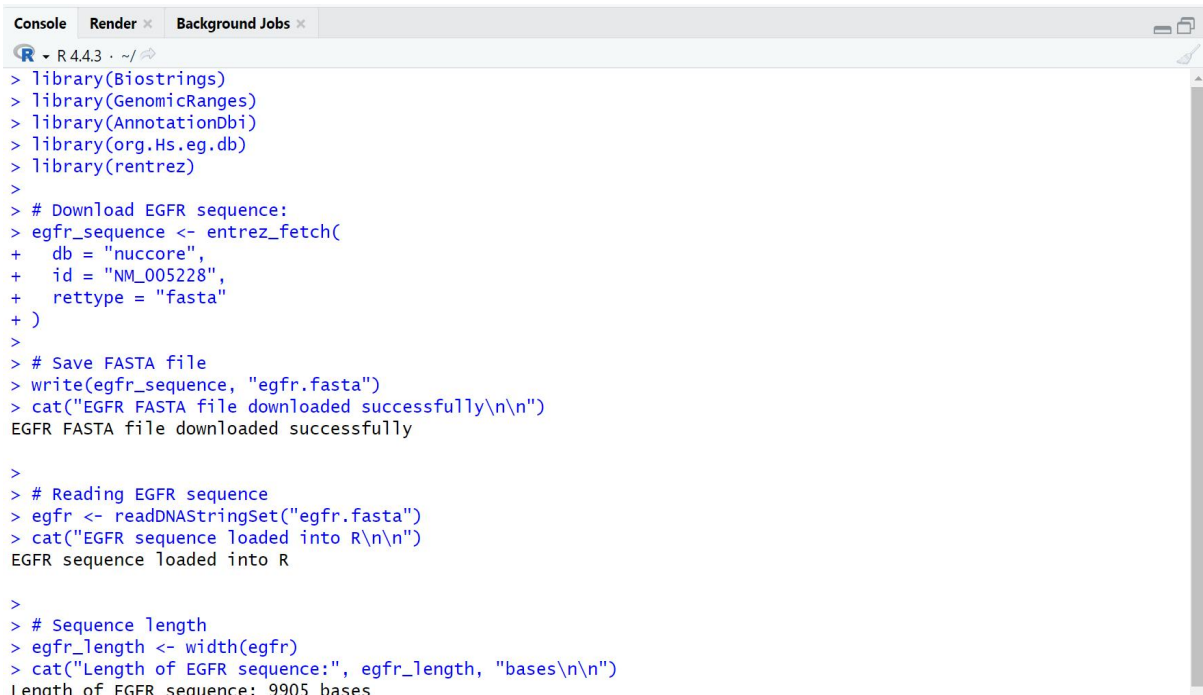
# Download EGFR sequence:
egfr_sequence <- entrez_fetch(
  db = "nuccore",
  id = "NM_005228",
  rettype = "fasta"
)

# Save FASTA file
write(egfr_sequence, "egfr.fasta")
cat("EGFR FASTA file downloaded successfully\n\n")

# Reading EGFR sequence
egfr <- readDNAStringSet("egfr.fasta")
cat("EGFR sequence loaded into R\n\n")

# Sequence length
egfr_length <- width(egfr)
```

**Output:**



```
R 4.4.3 · ~/
> library(Biostrings)
> library(GenomicRanges)
> library(AnnotationDbi)
> library(org.Hs.eg.db)
> library(rentrez)
>
> # Download EGFR sequence:
> egfr_sequence <- entrez_fetch(
+   db = "nuccore",
+   id = "NM_005228",
+   rettype = "fasta"
+ )
>
> # Save FASTA file
> write(egfr_sequence, "egfr.fasta")
> cat("EGFR FASTA file downloaded successfully\n\n")
EGFR FASTA file downloaded successfully

>
> # Reading EGFR sequence
> egfr <- readDNAStringSet("egfr.fasta")
> cat("EGFR sequence loaded into R\n\n")
EGFR sequence loaded into R

>
> # Sequence length
> egfr_length <- width(egfr)
> cat("Length of EGFR sequence:", egfr_length, "bases\n\n")
Length of EGFR sequence: 9905 bases
```

**Q2) What are the first 30 bases of the sequence?**

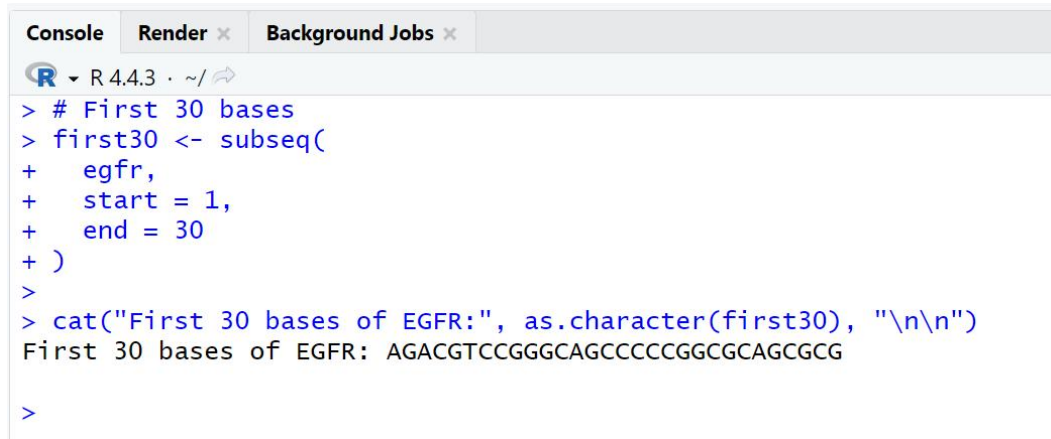
**Result:**

**Code:**

```
# First 30 bases
first30 <- subseq(
  egfr,
  start = 1,
  end = 30
)

cat("First 30 bases of EGFR:", as.character(first30), "\n\n")
```

**Output:**

A screenshot of an R console window showing the execution of R code. The console has tabs for 'Console', 'Render', and 'Background Jobs'. The code entered is: > # First 30 bases, > first30 <- subseq(, + egfr,, + start = 1,, + end = 30, + ), >, > cat("First 30 bases of EGFR:", as.character(first30), "\n\n"), First 30 bases of EGFR: AGACGTCCGGGCAGCCCCGGCGCAGCGCG, >. The output shows the first 30 bases of the EGFR sequence.

```
R 4.4.3 · ~/
> # First 30 bases
> first30 <- subseq(
+   egfr,
+   start = 1,
+   end = 30
+ )
>
> cat("First 30 bases of EGFR:", as.character(first30), "\n\n")
First 30 bases of EGFR: AGACGTCCGGGCAGCCCCGGCGCAGCGCG
>
```

**Q3) How many times does the pattern "ATG" appear in the sequence?**

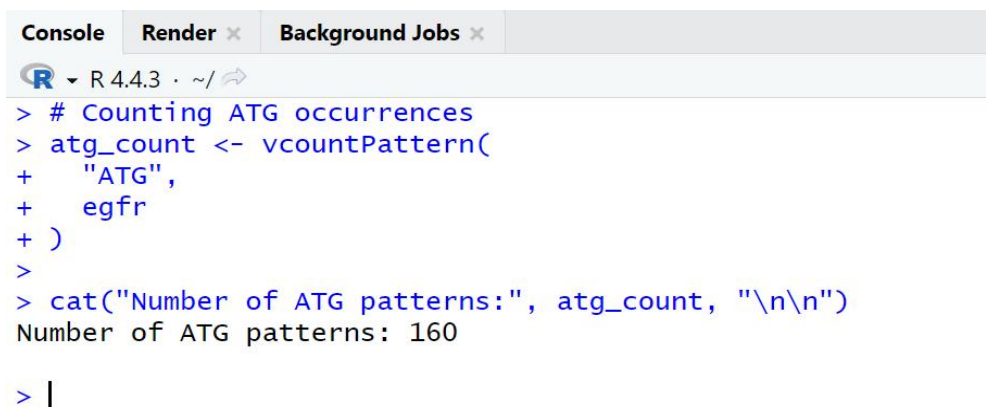
**Result:**

**Code:**

```
# Counting ATG occurrences
atg_count <- vcountPattern(
  "ATG",
  egfr
)

cat("Number of ATG patterns:", atg_count, "\n\n")
```

**Output**

A screenshot of an R console window showing the execution of R code. The console has tabs for 'Console', 'Render', and 'Background Jobs'. The code entered is: > # Counting ATG occurrences, > atg\_count <- vcountPattern(, + "ATG",, + egfr, + ), >, > cat("Number of ATG patterns:", atg\_count, "\n\n"), Number of ATG patterns: 160, > |. The output shows the number of ATG patterns in the sequence.

```
R 4.4.3 · ~/
> # Counting ATG occurrences
> atg_count <- vcountPattern(
+   "ATG",
+   egfr
+ )
>
> cat("Number of ATG patterns:", atg_count, "\n\n")
Number of ATG patterns: 160
> |
```

**Q4) What is the reverse complement of the first 30 bases?**

**Result:**

**Code:**

```
# Reverse complement
rev_comp <- reverseComplement(first30)
cat("Reverse complement of first 30 bases:", as.character(rev_comp), "\n\n")
```

Output:

```
Console Render x Background Jobs x
R 4.4.3 · ~/
> # Reverse complement
> rev_comp <- reverseComplement(first30)
> cat("Reverse complement of first 30 bases:", as.character(rev_comp), "\n\n")
Reverse complement of first 30 bases: CGCGCTGCGCCGGGGGCTGCCCGGACGTCT
```

Q5) Create a GRanges object for EGFR using the table above. What is its width (end – start)?

Result:

Code: # Creating GRanges object for EGFR

```
egfr_range <- GRanges(

  seqnames = "chr7",

  ranges = IRanges(
    start = 55019017,
    end = 55211628
  ),

  strand = "+"
)

print(egfr_range)

cat("\nGenomic width:", width(egfr_range), "bases\n\n")
```

Output:

```
Console Render x Background Jobs x
R 4.4.3 · ~/
>
> print(egfr_range)
GRanges object with 1 range and 0 metadata columns:
      seqnames      ranges strand
      <Rle>        <IRanges> <Rle>
[1] chr7 55019017–55211628      +
-----
seqinfo: 1 sequence from an unspecified genome; no seqlengths
>
> cat("\nGenomic width:", width(egfr_range), "bases\n\n")

Genomic width: 192612 bases

> |
```

Q6) Using AnnotationDbi and org.Hs.eg.db, what is the ENTREZ ID and full gene name for "EGFR"?

Result:

Code: # EGFR annotation

```
egfr_annotation <- select(
  org.Hs.eg.db,

  keys = "EGFR",

  columns = c(
    "ENTREZID",
    "GENENAME"
  ),

  keytype = "SYMBOL"
)

print(egfr_annotation)
```

Output:

```
Console Render x Background Jobs x
R 4.4.3 · ~/
+
+   columns = c(
+     "ENTREZID",
+     "GENENAME"
+   ),
+
+   keytype = "SYMBOL"
+ )
'select()' returned 1:1 mapping between keys and columns
>
> print(egfr_annotation)
  SYMBOL ENTREZID                                GENENAME
1    EGFR      1956 epidermal growth factor receptor
> |
```

**Q7) In 2–3 sentences, summarize what you learned about EGFR's sequence and location.**

**Result:** EGFR is located on the plus strand of chromosome 7, spanning 192,612 bases (55,019,017–55,211,628, GRCh38). The analyzed EGFR mRNA transcript (NM\_005228) was found to be 9,905 bases long and contained 160 ATG occurrences. EGFR encodes a receptor tyrosine kinase involved in cell signaling and is an important gene associated with cancer biology and targeted therapeutic approaches.